Status	Finished
Started	Friday, 30 October 2026, 2:51 PM
Completed	Friday, 30 October 2026, 2:54 PM
Duration	2 mins 47 secs
Grade	**11.00** out of 15.00 (**73.33**%)

Question **1**

Complete

Mark 0.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort

b. Heap sort

c. Merge sort

d. Selection sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach

b. It does not require recursion

c. It sorts data using a single pass

d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Linear Search
- c. Merge Sort
- d. Binary Search

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 0.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Bubble Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It makes the analysis slower
- c. It has no effect on the analysis process
- d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Bubble sort

b. Quick sort

c. Insertion sort

d. Heap sort

The correct answer is:
Insertion sort

Question **7**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting

b. Dividing the input into parts, solving each part, and combining the solutions

c. Sorting data sequentially

d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Dynamic programming

b. Backtracking

c. Divide and conquer

d. Greedy method

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting helps in finding duplicates quickly
c. Sorting is rarely used in programming
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **10**

Complete

Mark 0.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Bubbling up the largest element to its correct position
c. Dividing the list into halves
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. arrange()
c. order()
d. sort()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are discarded
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Being the most efficient sorting algorithm
c. Bubbling up the largest element to its correct position with each pass
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It divides the list into two halves, sorts each half, and then merges them
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It builds a sorted array one element at a time
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them